ets_marl - Sweep Analysis

April 30, 2026

1 ETS MARL — Sweep Analysis

Counter-part to `scripts/sweep.py`. This notebook is **analysis-only** — it does not train or instantiate the environment. It loads the CSV logs produced by a sweep run (typically on Azure) and supports two complementary modes of investigation:

1. **Top-level comparison across all variants and seeds** — price, reward, investment, compliance, secondary market, and newer diagnostics over training.
2. **Single-run deep-dive** — pick one `(variant, seed)` and reproduce the detailed D / A analysis from `Full Run & Analysis.ipynb` for that single run, with the green-investing and collateral-only sections removed and collateral folded into the reward decomposition panel.

1.1 1. Setup — Locate Project Root

No training, no environment construction. We only need access to `src/` for a few helpers (paths, optional eval) and to the `configs/` and `results/` folders.

```
Working directory: C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-
Energy-Auction
Config path: C:\Users\danie\Documents\Python_Scripts\Master Thesis\Thesis-
Energy-Auction\configs\default.yaml
```

1.2 2. Imports

1.3 3. Load the Sweep Specification

Point `SWEEP_SPEC` at the YAML you launched the sweep with. The sweep launcher (`scripts/sweep.py`, see `src/utils/sweep.py` for schema) deep-merges each variant's `overrides` onto the `base_config` and writes results under `<output_dir>/<variant_name>/`, with filenames tagged by `--run-tag <variant>` to keep them unique.

```
Sweep spec        : C:\Users\danie\Documents\Python_Scripts\Master
Thesis\Thesis-Energy-Auction\configs\sweeps\default_vs_unsold_to_msr.yaml
Base config       : configs/default.yaml
Output dir        : C:\Users\danie\Documents\Python_Scripts\Master
Thesis\Thesis-Energy-Auction\results\sweeps   (exists=True)
Default seeds     : []
parallel_workers  : 4
threads_per_worker: None
#variants         : 2
```

1.3.1 3a. Variant matrix

Flatten each variant's overrides into a single row so the differences are immediately visible.

```
            variant  n_seeds     seeds ets.unsold_to_msr
0           default        2  [23, 41]                 -
1  unsold_to_msr_on        2  [23, 53]              True
```

1.4 4. Discover Runs

For every `(variant, seed)` pair we look for the per-episode CSV and the per-year CSV. The sweep launcher writes them with a `--run-tag <variant>` infix; older single-config runs and locally-run smoke tests do not. We try the tagged filename first and fall back to the untagged one.

```
Found any log for 4 / 4 (variant, seed) pairs.
  training logs: 4
  year logs    : 4

            variant  seed                                        variant_dir
 training_log                                          year_log
 has_training_log  \
0           default    23  C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…              True
1           default    41  C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…              True
2  unsold_to_msr_on    23  C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…              True
3  unsold_to_msr_on    53  C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…
C:\Users\danie\Documents\Python_Scripts\Master…              True

   has_year_log  found
0          True   True
1          True   True
2          True   True
3          True   True
```

1.4.1 4a. Load all available runs into memory

We also resolve each variant's effective config (base overrides) so deep-dive cells can read e.g. `companies.n_agents`.

```
Loaded 4 runs covering 2 variants.
  complete   (train+year): 4
  train-only             : 0
  year-only              : 0
```

```
default/s 23 [TY] - 120000 eps, 1440000 year-rows
default/s 41 [TY] - 120000 eps, 1440000 year-rows
unsold_to_msr_on/s 23 [TY] - 120000 eps, 1440000 year-rows
unsold_to_msr_on/s 53 [TY] - 120000 eps, 1440000 year-rows
```

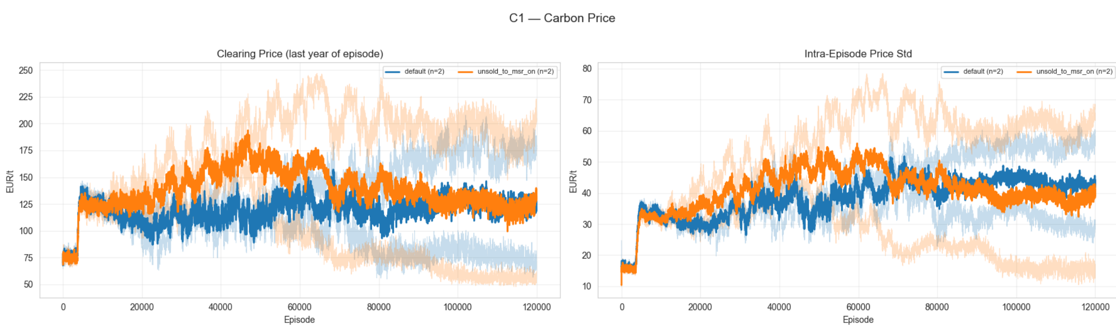
1.5 5. Top-Level Comparison Across Runs

All comparison plots use the same pattern:

1. Compute a per-episode scalar metric for every loaded run.
2. Smooth with a rolling mean (`COMPARE_ROLL`) — eyeballed at ~1% of `n_episodes` so different sweep sizes look comparable.
3. Plot one thin line per seed (faded) and one bold line per variant (mean across seeds), so within-variant noise vs. between-variant differences is visible at a glance.

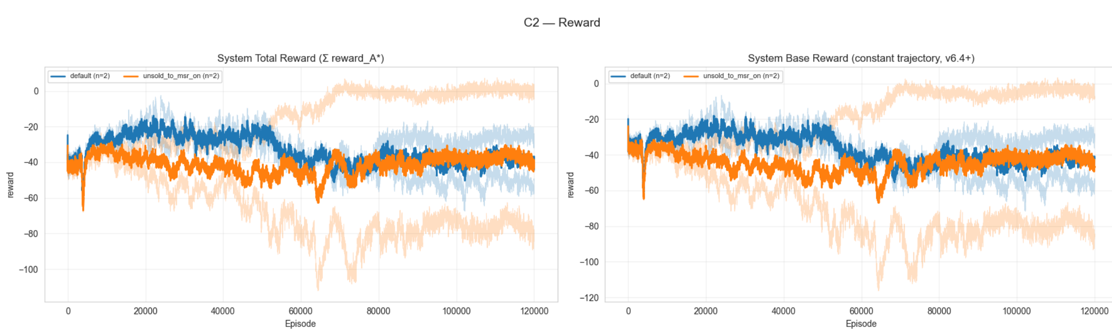
1.5.1 5.1 Carbon Price Over Training

Last-year clearing price per episode. The single number that summarises whether scarcity is biting.



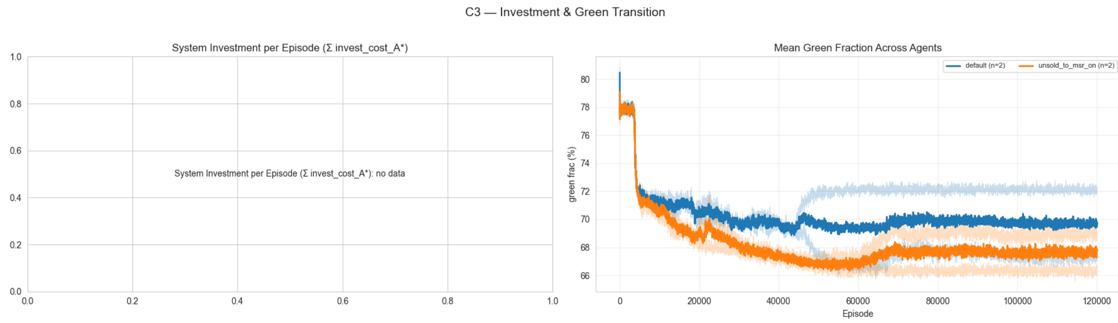
1.5.2 5.2 Reward Over Training

System reward (sum across learning agents) per episode. If `reward_base_A*` columns exist (v6.4+), we plot the constant-trajectory base reward separately from the decaying shaping bonus.



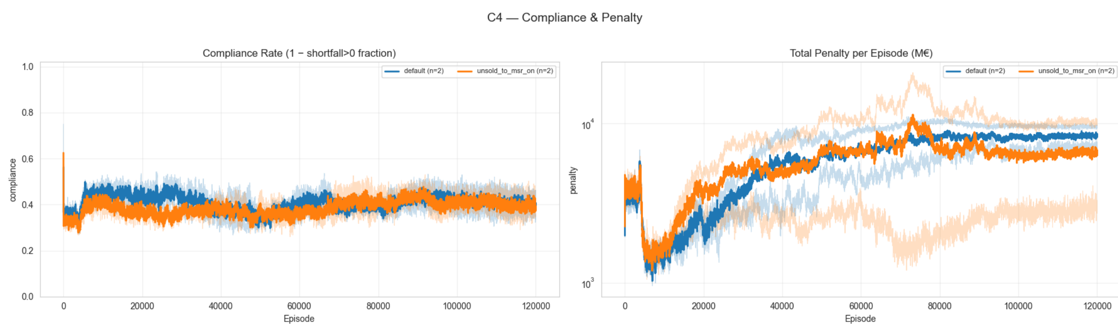
1.5.3 5.3 Investment Over Training

Total invested capital per episode and the system-mean green fraction.



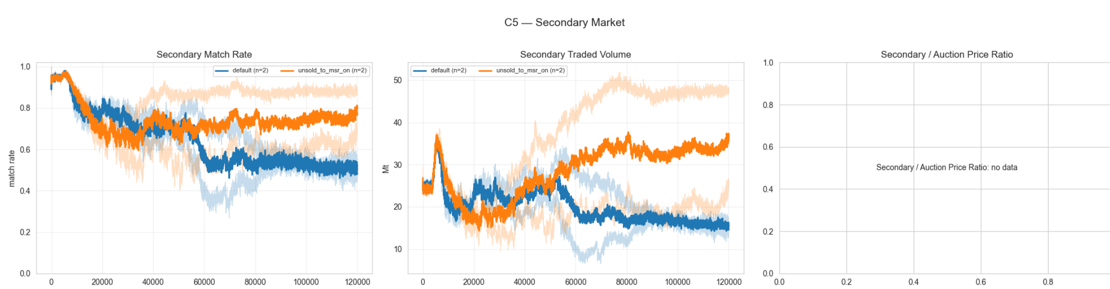
1.5.4 5.4 Compliance Over Training

Compliance rate = fraction of `(agent, episode)` cells with `shortfall_A*` 0. Total system penalty per episode is the second panel.



1.5.5 5.5 Secondary Market Over Training

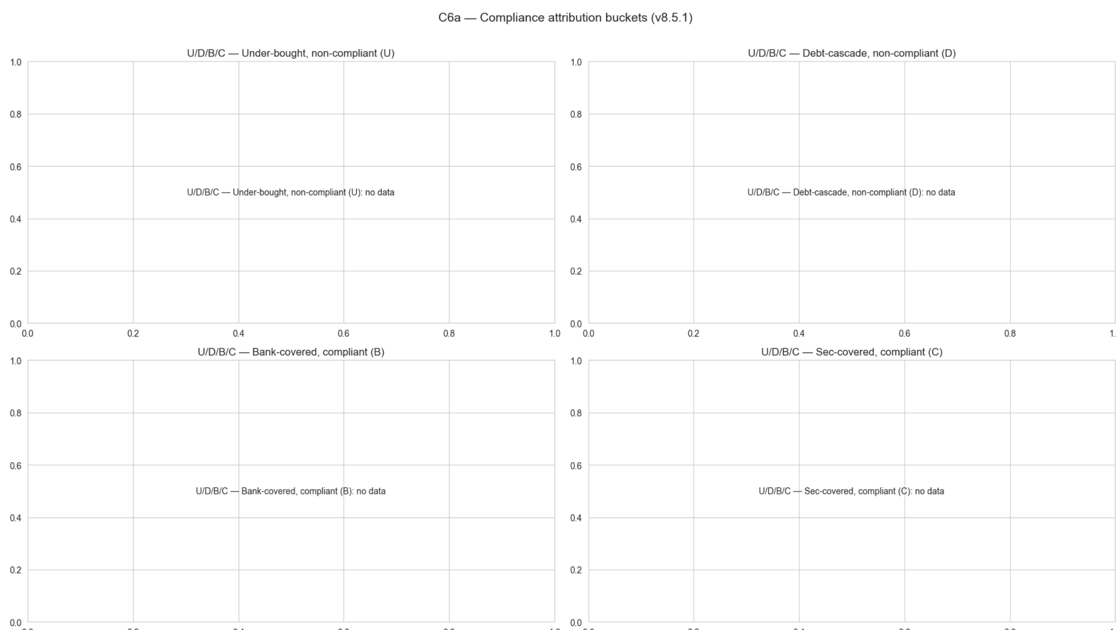
Match rate, traded volume, and the secondary-vs-auction price ratio. The ratio is informative for spotting price-spike regimes (sec auction) where the v8.5.2/v8.5.3 `expected_remediation_rate_real` proxy kicks in.

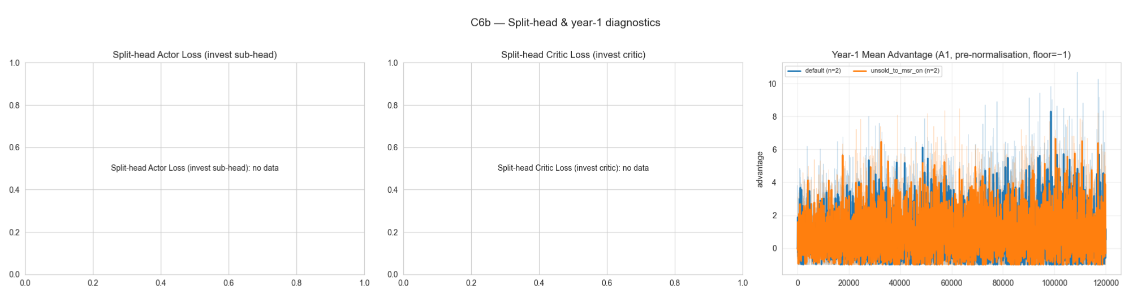


1.5.6 5.6 Newer Diagnostics (added since the original notebook)

These columns / fields are written by recent `train.py` versions; older logs simply lack them and the relevant panels degrade gracefully.

- **U / D / B / C compliance attribution** (v8.5.1) — partitions every `(agent, year)` into mutually exclusive buckets so you can tell whether non-compliance was driven by under-bidding (`U`) or by a debt cascade from prior carry-forward (`D`), and whether compliance came from the bank (`B`) or from the secondary market (`C`).
- **Year-1 advantage clamp diagnostic** (v8.4.1) — `adv_yr1_mean_A1` exposes the pre-normalisation year-1 GAE advantage, which the trainer now floors at -1.0 to stop early-training year-1 transitions from blowing out policy gradients.
- **Split-head investment critic losses** (v8.5) — when `ppo.split_invest_head=true`, separate `actor_loss_invest_A*` and `critic_loss_invest_A*` columns appear in the episode CSV.





1.5.7 5.7 Per-Variant Summary Table (Converged Window)

All metrics averaged over the last `ANALYSIS_TAIL_FRAC` of episodes — a single row per variant for thesis tables.

```
Per-(variant, seed) tail-window summary:

           variant  seed  price_mean  price_std  sec_match_rate  system_reward ␣
↪compliance_rate  system_penalty  system_invest  green_frac_mean ␣
↪collateral_year_mean  n_episodes
0          default    23     171.257     49.244           0.482        -49.821 ␣
↪           0.365        7199.304            NaN           0.673               ␣
↪55.480      120000
1          default    41      74.943     38.317           0.540        -25.489 ␣
↪           0.452        9547.589            NaN           0.721               ␣
↪32.884      120000
2  unsold_to_msr_on    23      56.357     25.055           0.881          0.690 ␣
↪           0.438        2895.218            NaN           0.690               ␣
↪32.224      120000
3  unsold_to_msr_on    53     187.861     61.202           0.625        -75.113 ␣
↪           0.362       10217.163            NaN           0.663               ␣
↪56.004      120000

Aggregated by variant (mean ± std across seeds):

                  price_mean_mean  price_mean_std  system_reward_mean
system_reward_std  compliance_rate_mean  compliance_rate_std  system_invest_mean
system_invest_std  green_frac_mean_mean  \
variant
default                   123.100          68.104             -37.655
17.206                0.409               0.062                 NaN
NaN                 0.697
unsold_to_msr_on          122.109          92.988             -37.211
53.601                0.400               0.054                 NaN
NaN                 0.676

                  green_frac_mean_std  collateral_year_mean_mean
sec_match_rate_mean  system_penalty_mean  n_seeds
variant
default                         0.034                     44.182
0.511                8373.446         2
unsold_to_msr_on                0.019                     44.114
0.753                6556.190         2
```

1.6 6. Deep Dive — Pick One Run

Set `PICK_VARIANT` and `PICK_SEED` to the run you want to investigate in detail. Re-run the cells from here to refresh the deep-dive plots without recomputing the comparison sections above.

```
Deep-diving on: variant=unsold_to_msr_on, seed=53
  120000 episodes, 1440000 year-rows, 12 years/episode, 8 learning agents
  Converged window: last 200 episodes
```

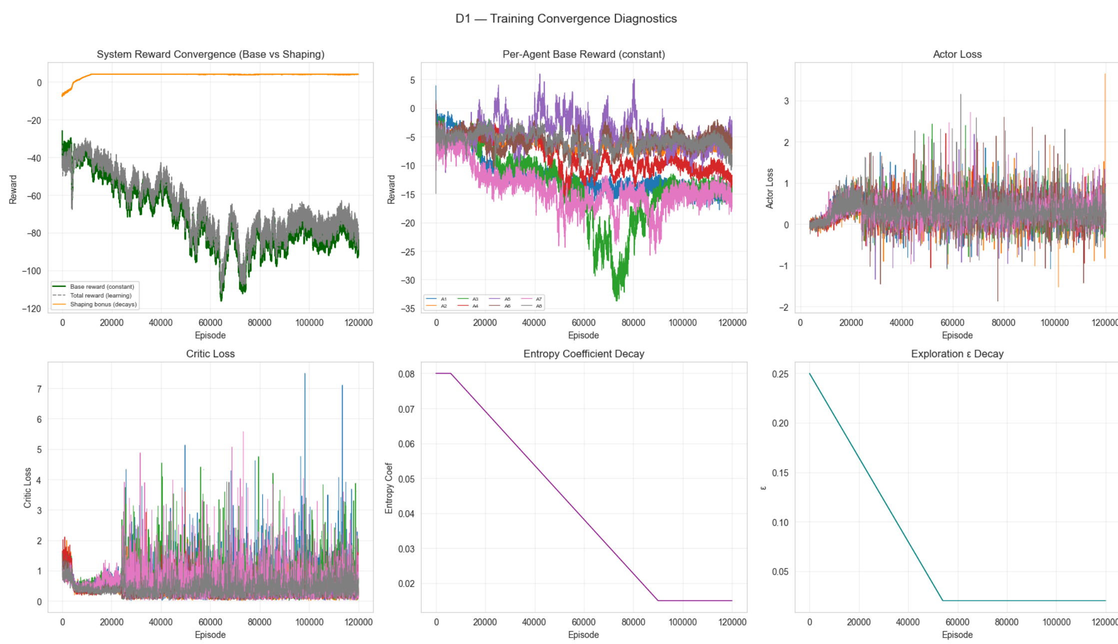
---

2 Part I — Diagnostic Deep Dive (D)

> *Is the simulation behaving correctly?*
>
> Reordered vs. the original notebook: **D5 (stochastic shocks) is last**, and the dedicated green-investing (D7) and collateral (D9 / D9b) sections are removed. Collateral is folded into the reward-decomposition panel in §A5.

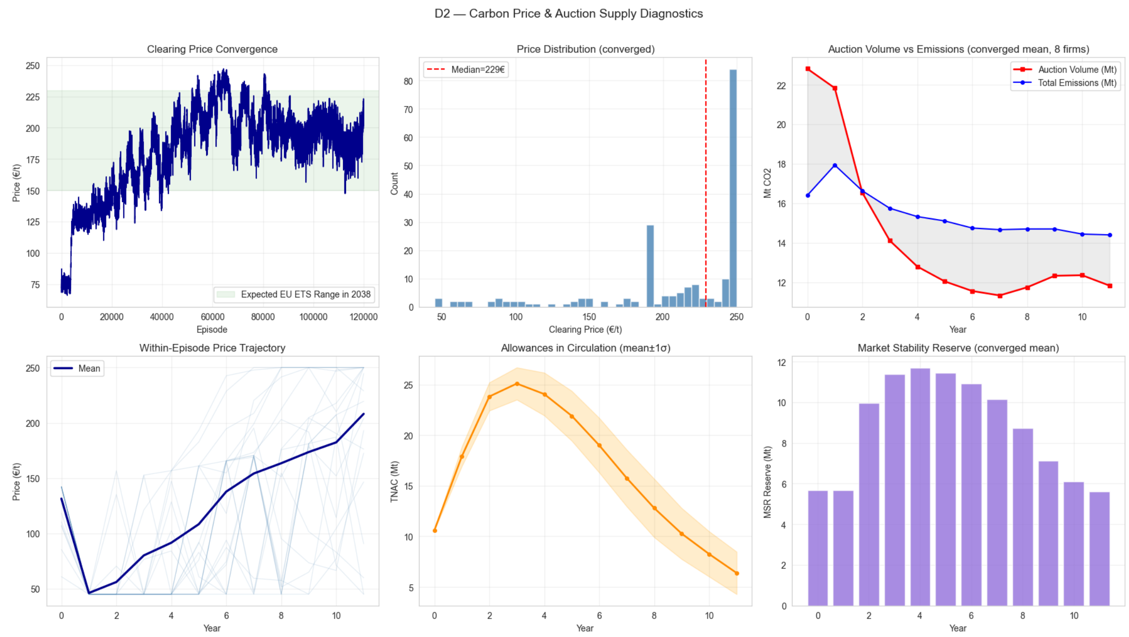
2.0.1 D1. Training Convergence

Check that neural networks are learning stably: - Losses decreasing - Entropy decay triggered at the right time - Rewards stabilizing



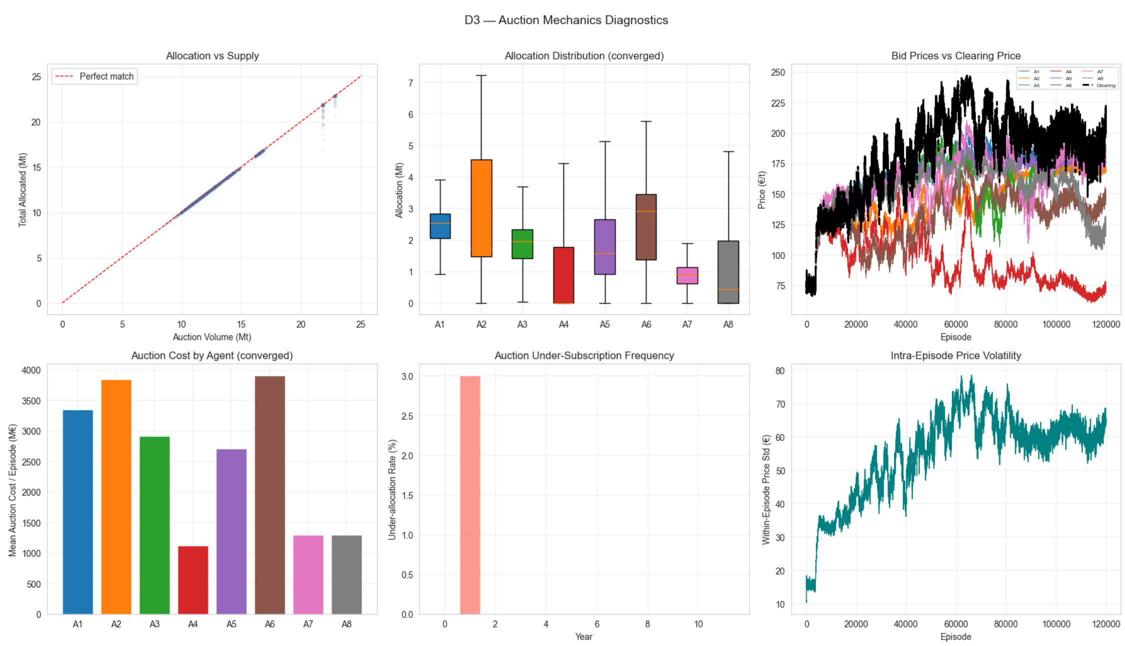
2.0.2 D2. Carbon Price & Cap Schedule

The clearing price should broadly reflect real EU ETS dynamics: - Phase 4 prices in the €50–100 range - Prices rising as the cap tightens under the LRF - TNAC declining over time as MSR absorbs surplus



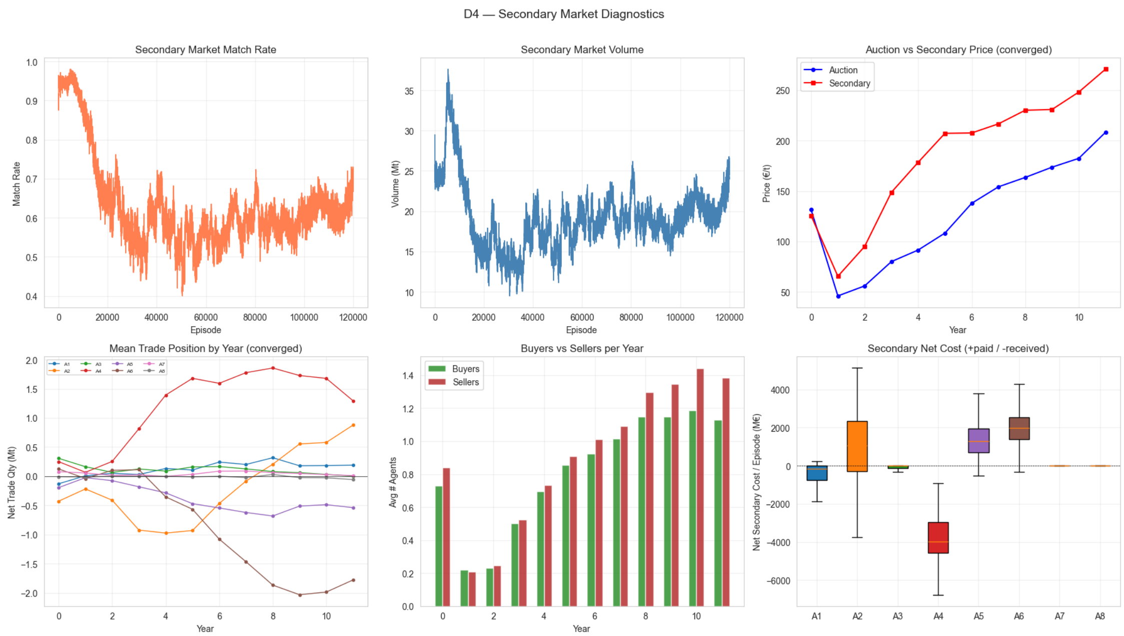
2.0.3 D3. Auction Mechanics

Sanity checks for the uniform-price auction: - Allocations should sum to roughly the auctioned volume - Bid prices should bracket the clearing price - No persistent auction failures



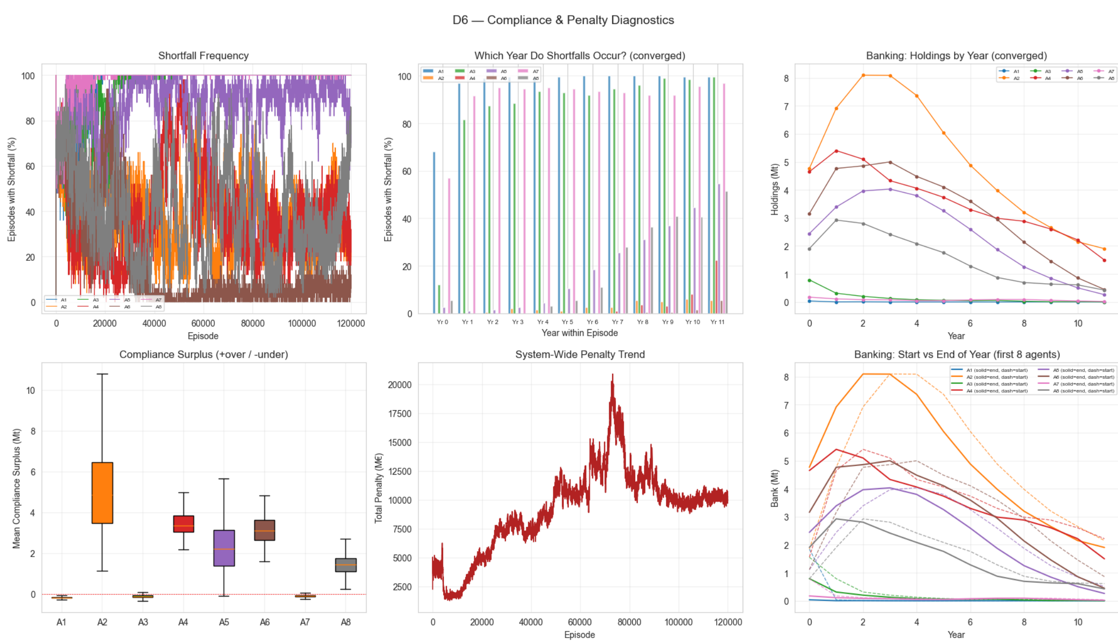
2.0.4 D4. Secondary Market Mechanics

Does the double-auction secondary market function properly? - Reasonable match rates (not all-or-nothing) - Secondary prices tracking auction prices - Both buyers and sellers participating



2.0.5 D6. Compliance & Penalty Mechanics

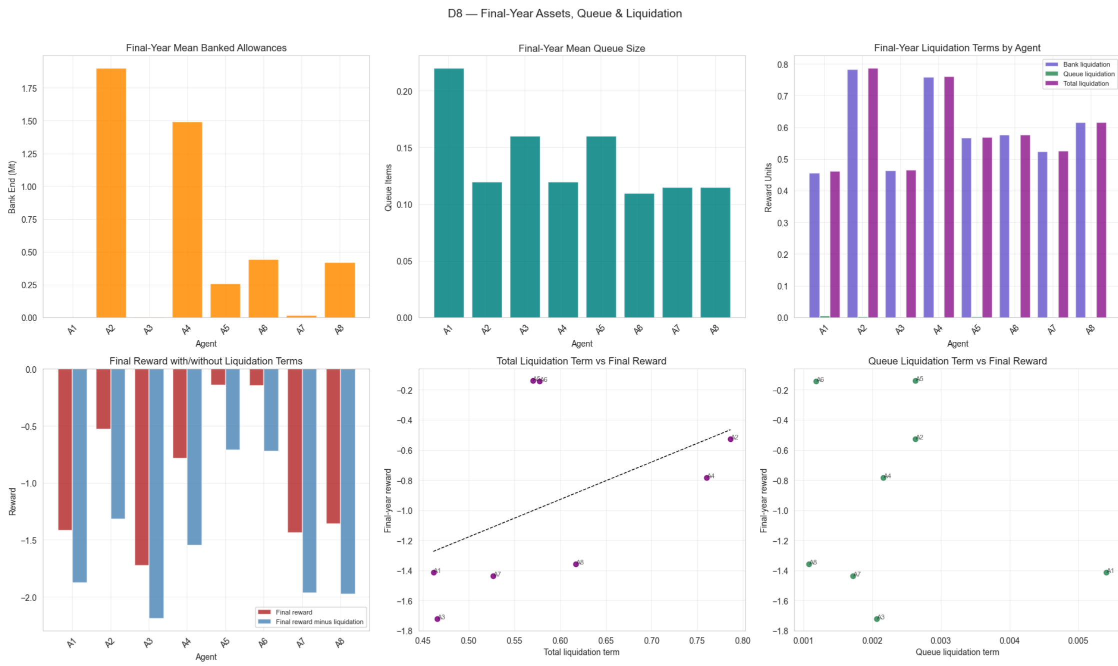
Verify that compliance mechanics work: - Shortfalls are detected and penalized - Penalties are proportional to shortfall - Banking (carry-over) functions correctly



2.0.6 D8. Final-Year Invested Assets, Queue State, and Liquidation Payoff

This cell now prioritizes liquidation values logged directly from the environment reward computation into year_log CSV.

Fallback behavior: - If `terminal_*_value_A*` columns are absent (older logs), bank liquidation is reconstructed from bank_end × clearing_price / 1000. - Queue liquidation is shown only if logged columns exist.



2.0.7 D5. Stochastic Shocks (P5 & P6)

Verify that emission demand shocks (P5) and construction uncertainty (P6) are active and producing the intended distributions.


```
P5 emission shock active: True
P6 CF-noise active:       True
P6 cancellation active:   True
Inflation logged:         True
Inflation mean:           2.005%
Inflation std:            1.488%-pt
```

---

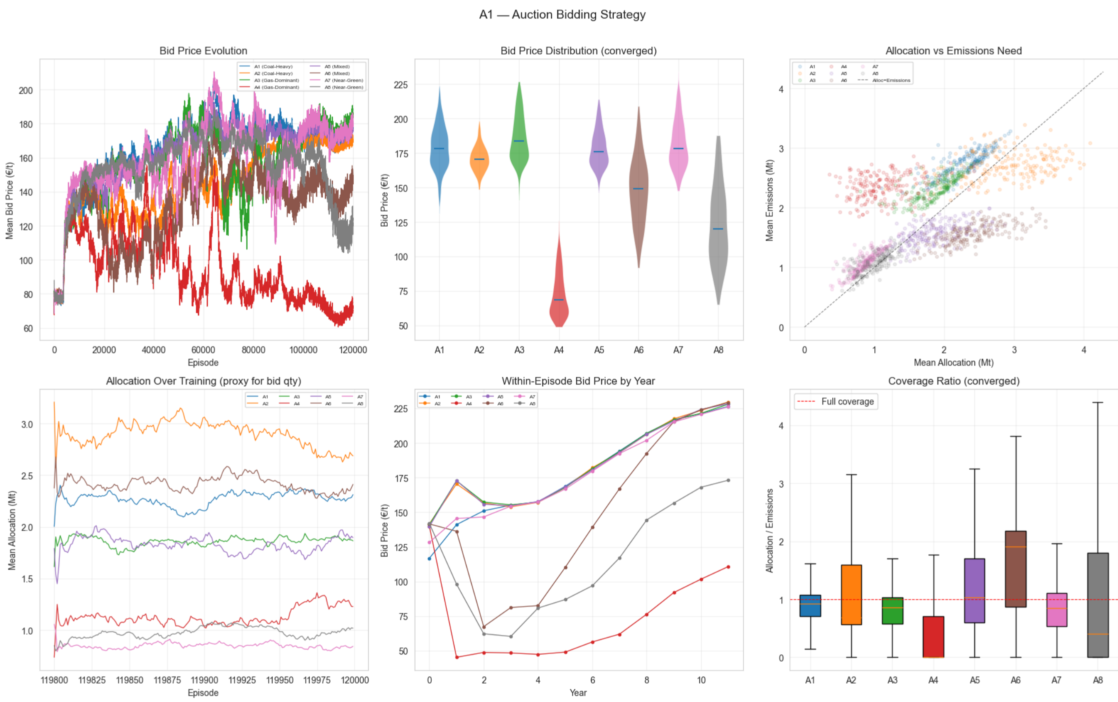
3 Part II — Strategy Deep Dive (A)

What strategies did the agents learn and why?

Drops the bot-comparison section (A7 / A7a) and the standalone summary / troubleshooting digest. The standalone collateral panel (A5b) is removed; collateral is folded into the cost-breakdown panel of §A5.

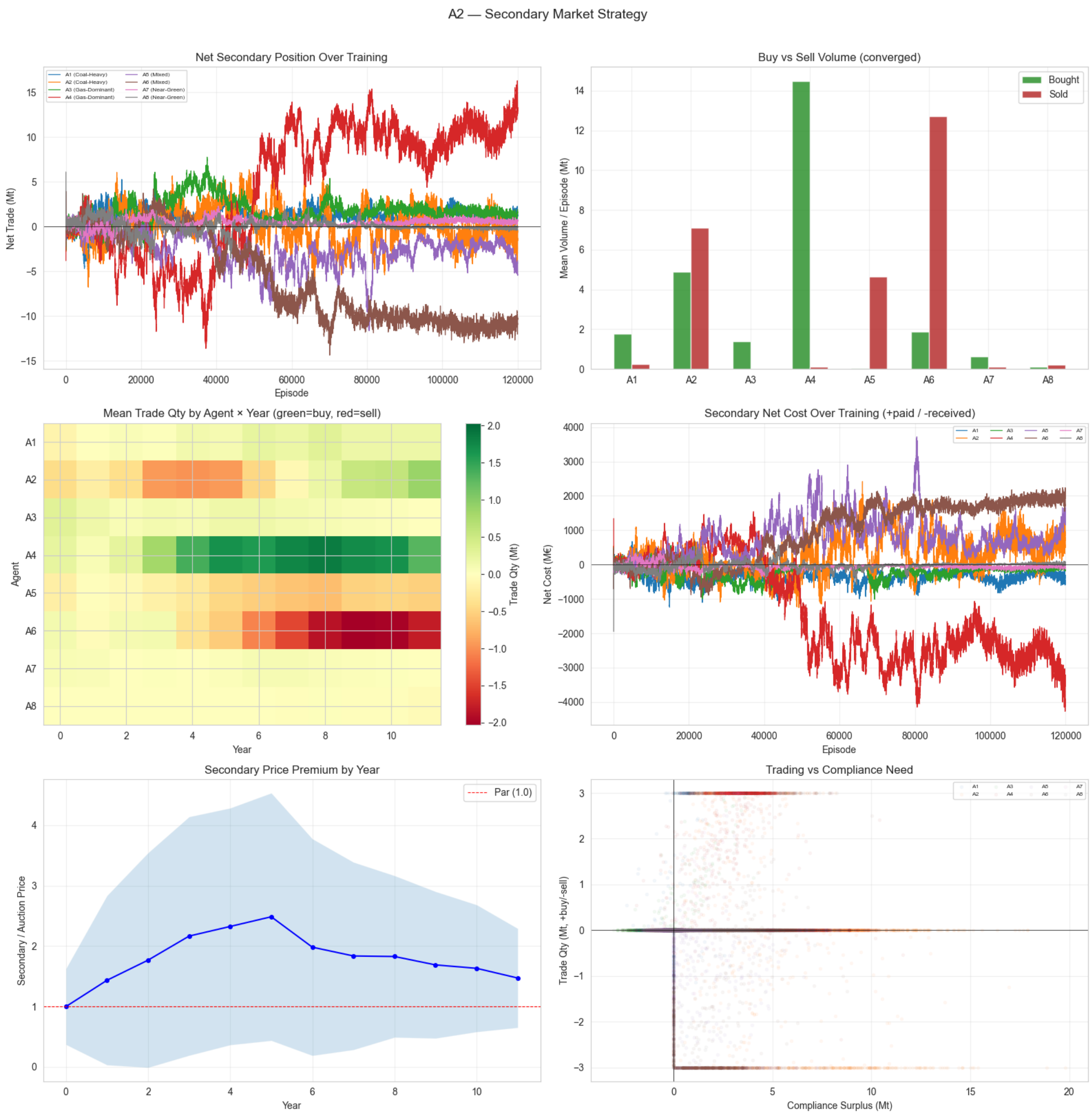
3.0.1 A1. Auction Bidding Strategy

How agents bid in the primary EUA auction: - **Price discovery** — do bids converge to a common clearing price? - **Quantity strategy** — do agents bid for what they need or over/under-bid? - **Archetype differences** — do coal-heavy agents bid differently from near-green?



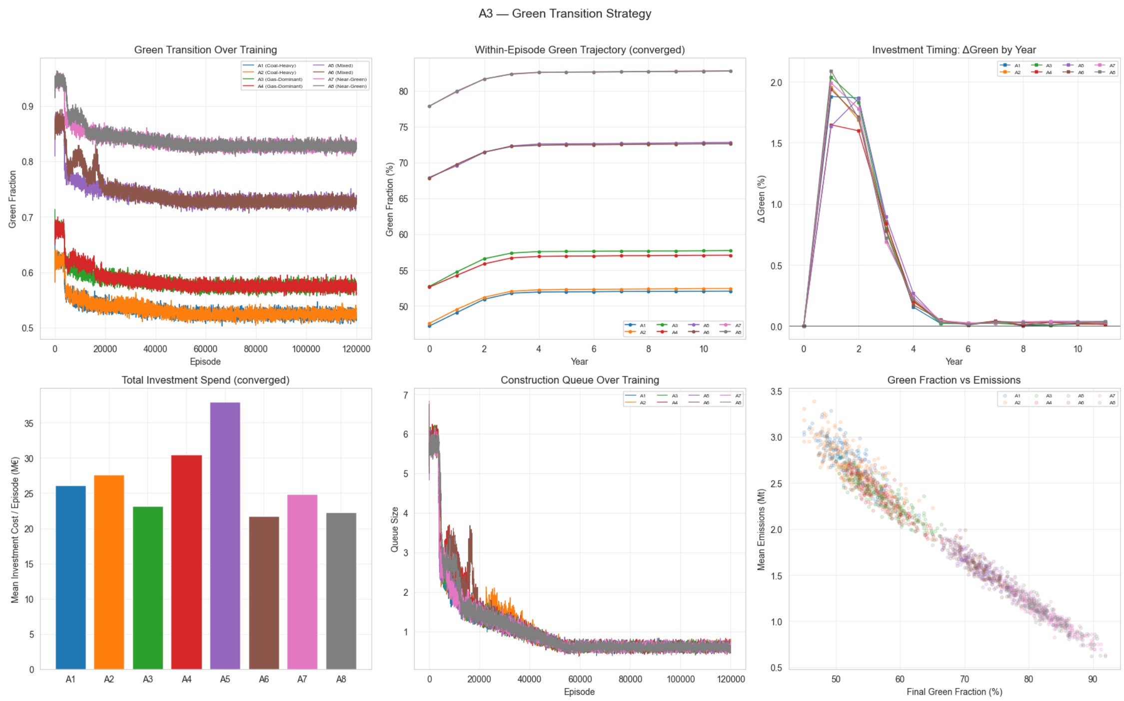
3.0.2 A2. Secondary Market Strategy

How agents trade on the secondary market: - **Role evolution** — who becomes a net buyer vs seller over training? - **Year-by-year patterns** — do roles shift within an episode as the cap tightens? - **Price premiums** — secondary vs auction price spread - **Cost/revenue** — who profits from secondary trading?



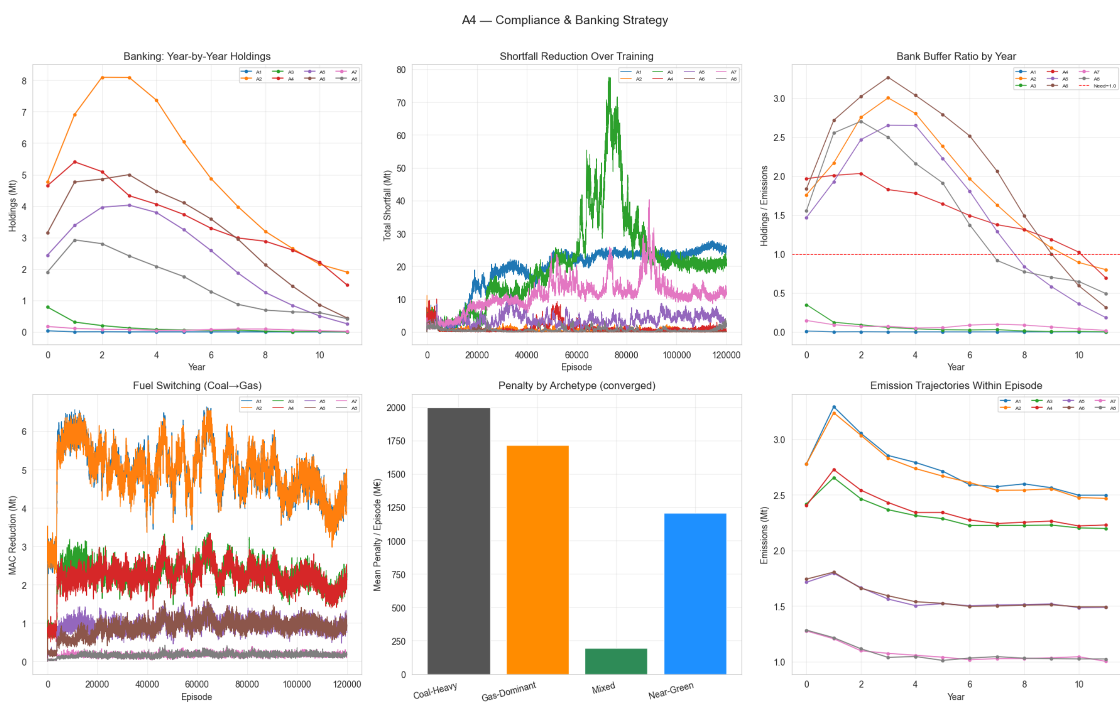
3.0.3 A3. Green Transition Strategy

Investment patterns in renewable energy: - **Transition speed** — how quickly does each archetype green its portfolio? - **Investment intensity** — how aggressively do agents invest? - **Technology preference** — onshore wind, offshore wind, or solar? - **Construction pipeline** — queuing effects and timing



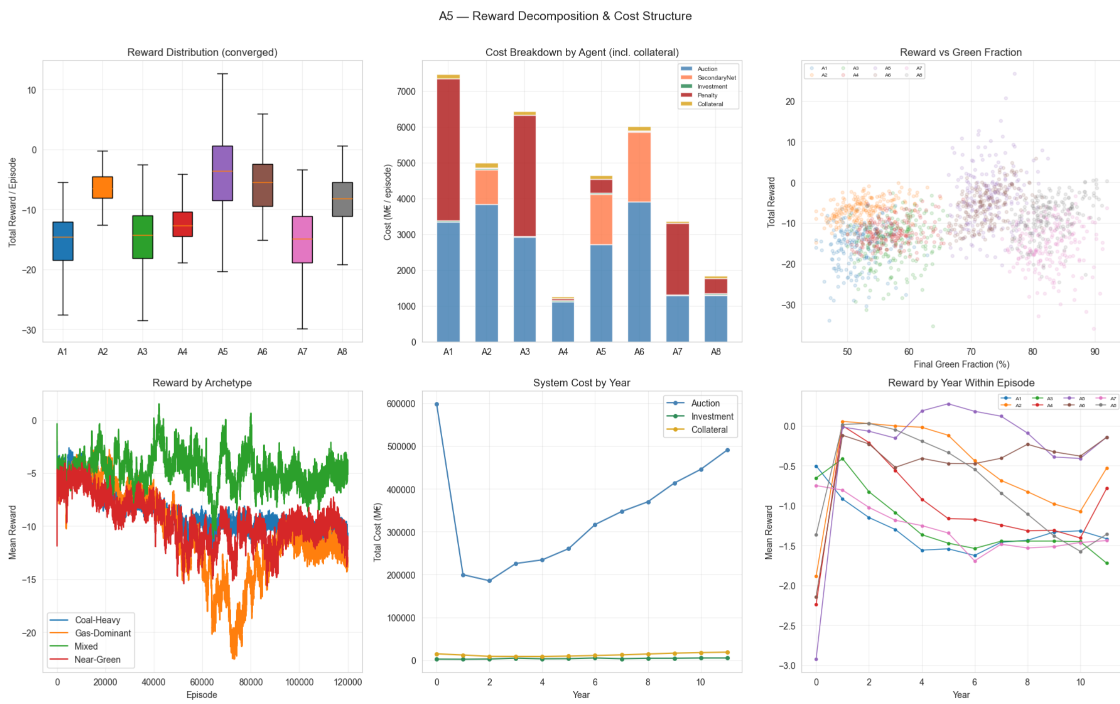
3.0.4 A4. Compliance & Banking Strategy

How agents manage their allowance portfolio: - **Banking** — accumulating surplus for future compliance - **Shortfall avoidance** — learning to stay compliant - **Risk management** — holdings buffer vs cost minimization



3.0.5 A5. Reward Decomposition & Cost Structure

Decompose the per-agent reward into its main cost components. The cost-breakdown panel now folds in **collateral cost** alongside the four original components (auction, secondary, investment, penalty). Older logs without `collateral_cost_A*` columns simply skip that layer.



3.0.6 A6. Deterministic Evaluation of Best Agents (optional)

Loads the best checkpoint of each agent for the chosen `(variant, seed)` and runs one deterministic episode. Skip with `RUN_A6_EVAL = False` if the sweep was run on a remote machine and only CSV logs are available locally — A6 is the only deep-dive cell that needs the actual environment.

```
A6 skipped (RUN_A6_EVAL=False).
```